

Team Members:

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Tentative Title of Project:

Mushroom Kingdom: Image-Based Classification and Clustering of Mushroom Species

Dataset link:**Combined Kaggle Mushroom Images Dataset**

<https://www.kaggle.com/dariobaumberger/combined-kaggle-mushrooms-dataset/data>

The Dataset above was combined from the following 2:

<https://www.kaggle.com/zlatan599/mushroom1>

<https://www.kaggle.com/thehir0/mushroom-species>

This dataset is a merged collection of several Kaggle mushroom image datasets, comprising over 100,000 images spanning more than 100 mushroom species identified by Latin names

Research questions:

- Can CNN accurately classify mushroom species from images?
- How do mushroom species cluster based on learned image features?
- What biological or practical characteristics do these clusters correspond to?
- Can unsupervised clustering identify mislabeled observations or outlier images within the dataset?

Description of summary statistics and methods we plan to use:

- Exploratory Data Analysis (EDA): Examine class distributions across species and image dimensions. Identify and handle class imbalances if present.
- Train/Validation/Test Split: Use a 70/15/15 split with stratification to preserve class proportions across sets.
- Preprocessing: will include image resizing if needed, normalization of pixel values, and data augmentation to improve generalization.
- CNN Classification: Train a CNN on the image dataset. Evaluate using accuracy, precision, recall, and F1-score per class.
- Feature Extraction and Clustering: Hierarchical clustering, dendrogram generation.
- Data Cleaning: Detection of potential mislabeled images or outliers, dataset refinement, followed by retraining